① The email address of the Discourse admin must be valid.

② Creates a VPC in the address range 172.31.0.0/16

③ We want to access Discourse from the internet, so we need an internet gateway.

④ Attaches the internet gateway to the VPC

⑤ Creates a subnet in the address range 172.31.38.0/24 in the first availability zone (array index 0)

⑥ Creates a second subnet in the address range 172.31.37.0/24 in the second availability zone (properties omitted)

⑦ Creates a route table that contains the default route, which routes all subnets in a VPC

⑧ Associates the first subnet with the route table

⑨ Adds a route to the internet via the internet gateway

The following listing adds the required network access control list.

Listing 11.3 CloudFormation template for Discourse: VPC NACLs

```
Resources:
  # [...]
  NetworkAcl:                          ①
    Type: AWS::EC2::NetworkAcl
    Properties:
      VpcId: !Ref VPC
  SubnetNetworkAclAssociationA:        ②
    Type: 'AWS::EC2::SubnetNetworkAclAssociation'
    Properties:
      SubnetId: !Ref SubnetA
      NetworkAclId: !Ref NetworkAcl
  # [...]
  NetworkAclEntryIngress:              ③
    Type: 'AWS::EC2::NetworkAclEntry'
    Properties:
      NetworkAclId: !Ref NetworkAcl
      RuleNumber: 100
      Protocol: -1
      RuleAction: allow
      Egress: false
      CidrBlock: '0.0.0.0/0'
  NetworkAclEntryEgress:               ④
    Type: 'AWS::EC2::NetworkAclEntry'
    Properties:
      NetworkAclId: !Ref NetworkAcl
      RuleNumber: 100
      Protocol: -1
      RuleAction: allow
      Egress: true
      CidrBlock: '0.0.0.0/0'
```

① Creates an empty network ACL

② Associates the first subnet with the network ACL

③ Allows all incoming traffic on the network ACL. (You will use security groups later as a firewall.)

④ Allows all outgoing traffic on the network ACL